Moving the Slots, Not the Balls: Conformal Prediction as the Contragredient of Herding

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Abstract

Herding constructs a point set whose empirical feature averages track a target measure: it *moves the particles* against fixed test functionals, and when its weight sequence stays bounded it earns an $O(1/n)$ discrepancy where i.i.d. sampling gives $O(1/\sqrt{n})$. We observe that the coverage argument behind split conformal prediction is the dual operation. The realized scores are never touched; the proof transports the *designation* "this slot holds the test point" across the index set, and because the empirical quantile is a symmetric function of the scores, a deterministic counting identity holds: exactly $\lceil (1-\alpha)(n+1) \rceil$ of the $n+1$ possible placements of that designation point at covered scores. In the roulette image: herding moves the balls to balance a fixed wheel; conformal holds the balls fixed and rotates the slot labels under the pointer. We make the duality precise as a statement about the permutation action on score vectors and its contragredient action on coordinate functionals, and we use it to read the switch-coefficient analysis of Barber and Pananjady (2026) as index-space low-discrepancy accounting: deleting τ slots damages the rank count by at most τ, worth τ/n of coverage, and the only probabilistic price is the total-variation failure of rotation invariance, $\bar{\Psi}_\tau \le 2\beta(\tau)$. The linear $1/n$ rates on both sides are the signature of counting rather than concentration, which is why they beat the $\sqrt{\tau/n}$ obtained when the counting structure is abandoned for blocking and empirical processes. We are explicit about what the duality does not claim: conformal performs no active steering, and the probabilistic work is still done by exchangeability or its mixing surrogate.

1 Two mechanisms

Herding moves the balls. Fix a feature map $\phi : \mathcal{X} \to \mathcal{H}$ and a target expectation $\mu = \mathbb{E}_P\, \phi(X)$. Herding (Welling, 2009) generates points deterministically,

$$x_t \;=\; \arg\max_{x \in \mathcal{X}} \langle w_{t-1},\, \phi(x) \rangle, \qquad w_t \;=\; w_{t-1} + \mu - \phi(x_t), \tag{1}$$

so that the weight vector $w_t = w_0 + t\mu - \sum_{s \le t} \phi(x_s)$ is, up to initialization, t times the running discrepancy between the target mean and the empirical feature mean. If the dynamics keep $\|w_t\|$ bounded — guaranteed in finite dimension under mild conditions, and more delicately in a general RKHS (Chen et al., 2010; Bach et al., 2012) — then

$$\left\| \tfrac{1}{n} \textstyle\sum_{t=1}^{n} \phi(x_t) - \mu \right\| \;=\; \tfrac{\|w_n - w_0\|}{n} \;=\; O\!\left(\tfrac{1}{n}\right), \tag{2}$$

a quasi–Monte Carlo rate, against the $O(n^{-1/2})$ of i.i.d. sampling. The tests ϕ are held fixed; the *state-space locations* of the particles are what the algorithm manipulates. Bach et al. (2012) identify (1) with the Frank–Wolfe algorithm on the marginal polytope, which is where the boundedness conditions live; Chen et al. (2010) read it as greedy quasi–Monte Carlo.

Conformal moves the slots. Split conformal prediction (Vovk et al., 2005; Lei et al., 2018) computes conformity scores $S_1, \ldots, S_n$ on a calibration set, a score S_{n+1} on the test point, and declares coverage when S_{n+1} falls below the inflated empirical quantile $Q_{1-\alpha}$ = the k-th smallest of $S_1, \ldots, S_n$, with $k = \lceil (1-\alpha)(n+1) \rceil$. Nothing in the procedure moves a score. The proof of validity instead moves a *label.* Write $S = (S_1, \ldots, S_{n+1}) \in \mathbb{R}^{n+1}$ for the realized scores — the balls, resting where they fell — and let e_i^* denote the coordinate functional that reads slot i. The coverage event is a statement about the pairing $\langle e_{n+1}^*, S \rangle$ and the symmetric function $Q_{1-\alpha}$ of the remaining coordinates. The proof asks: what if the designation "test" had pointed at slot i instead?

Observation 1 (The rank count is already balanced)**.** *Let $s_1, \ldots, s_{n+1}$ be distinct reals and $k = \lceil (1-\alpha)(n+1) \rceil$. For each placement $i \in \{1, \ldots, n+1\}$, say the placement is* covered *when s_i is at most the k-th smallest of the other n values. Then the number of covered placements is exactly k: placement i is covered if and only if the rank of s_i among all $n+1$ values is at most k. In particular the covered fraction is $k/(n+1) \geq 1-\alpha$,* deterministically, *for any score vector whatsoever.*

This is the whole finite-sample content of the guarantee. No mass is transported, no point is steered; the balance that herding must manufacture by moving particles is already present in the rank structure, and the proof merely rotates the pointer to exhibit it. Probability enters once, at the end: exchangeability says the realized designation (slot $n+1$) is indistinguishable from a uniform draw over the $n+1$ placements, so the deterministic fraction becomes a probability.

2 The contragredient

The duality can be said precisely. The symmetric group acts on score vectors through permutation matrices, $\rho(\pi)S = P_\pi S$; the contragredient (dual) representation acts on covectors, $\rho^*(\pi)f = f \circ \rho(\pi^{-1})$, so that the pairing is invariant:

$$\big\langle \rho^*(\pi)\, e_{n+1}^*,\, \rho(\pi)\, S \big\rangle = \big\langle e_{n+1}^*,\, S \big\rangle.$$

Every exchangeability proof of conformal validity can be run on either side of this pairing: *either* rearrange the data, $S \mapsto P_\pi S$, holding the test functional fixed, *or* hold the data fixed and pull the functional back, $e_{n+1}^* \mapsto \rho^*(\pi^{-1}) e_{n+1}^* = e_{\pi(n+1)}^*$. Because the quantile threshold is a symmetric — that is, invariant — function of the score multiset, the two descriptions agree event by event. The textbook proof is usually narrated in the first voice ("swap the test point with a calibration point"); Observation 1 is the second voice, and the second voice is the one in which the mechanism is visible: the orbit being averaged is the orbit of the *functional*, $\{e_i^*\}_{i=1}^{n+1}$, not an orbit of datasets.

Two honesty clauses. First, permutation representations are orthogonal, so the contragredient is *equivalent* to the original representation; the term is apt not because the dual carries different structure but because it names which side of the pairing the action sits on — the covector side. Second, herding and conformal are not the same algorithm run in opposite categories: herding actively optimizes (1), while conformal optimizes nothing. The duality lives at the level of the finite-sample ledger, and it is exact there:

herding moves the observations to balance the measure;
conformal moves the observation *labels* to expose an already-balanced rank count.

3 The $1/n$ ledger, and dependence as damaged slots

Both mechanisms produce $1/n$ rates, and for the same bookkeeping reason: an error that accumulates *boundedly* is divided by the sample size.

$$\text{herding:} \qquad \text{error} \;\le\; \frac{\text{bounded cumulative feature discrepancy } \|w_n\|}{n},$$
$$\text{conformal:} \qquad \text{coverage damage} \;\le\; \frac{\text{number of damaged or deleted ranks}}{n}.$$

The second line is not a metaphor; it is the proof technique of Barber and Pananjady (2026) for split conformal on a dependent series. There, exact slot symmetry is unavailable: rotating the test designation to slot k is no longer distribution-preserving, because neighbouring scores remember one another. The repair is to delete a buffer of τ slots around the splice before rotating. Deleting τ of n entries moves the k-th order statistic by at most τ ranks — order statistics are insertion/deletion stable — so the quantile level shifts by at most τ/n; and the residual failure of rotation invariance is charged at its total-variation price, the switch coefficient $\bar{\Psi}_\tau$, bounded by $2\beta(\tau)$ for a stationary β-mixing sequence. The coverage floor

$$\mathbb{P}(\text{cover}) \;\ge\; 1 - \alpha - \min_{\tau}\Big\{ \underbrace{\tfrac{\tau}{n+1}}_{\text{slots sacrificed}} + \underbrace{\bar{\Psi}_\tau}_{\text{failed invariance}} \Big\}$$

is index-space low-discrepancy accounting, and nothing in it concentrates: no variance appears, no square root can. The contrast is instructive — when the same problem is attacked by blocking and empirical-process concentration (Oliveira et al., 2024), the rate degrades to the Monte Carlo $\sqrt{\tau/n}$, precisely because the counting structure has been traded away for a maximal inequality. Herding's advantage over i.i.d. sampling and the switch coefficient's advantage over blocking are the same advantage.

The quasi–Monte Carlo reading closes the loop. Chen et al. (2010) present herding as a greedy low-discrepancy construction: choose the next point to keep the running discrepancy bounded. The rank sequence of any sample is, tautologically, a perfectly low-discrepancy sequence over its own index set — every rank appears exactly once — and Observation 1 is the Koksma–Hlawka-style consequence: a test functional averaged over slots inherits the balance of the ranks. Conformal prediction is what you get when the low-discrepancy sequence comes for free and only the pointer needs transporting. The quadrature reading of conformal prediction is not ours alone: Snell and Griffiths (2025) recast the coverage computation as Bayesian quadrature over the calibration scores, recovering the frequentist guarantee as a special case of a posterior over test loss. The slot-transport view is the deterministic, index-space face of the same coin: quadrature weights over scores on one side, a rotating evaluation functional over slots on the other.

4 What the duality does and does not explain

It explains the finite-sample ledger: why the guarantee is exact at finite n, why corrections enter linearly in the number of damaged slots, and why dependence is a tax of size (dependence range)/(calibration size) rather than its square root. It does not explain, and we do not claim it explains, the statistical content: the step from "k of $n+1$ placements are covered" to "the realized placement is covered with probability $k/(n+1)$" is exchangeability, or its quantitative surrogate under mixing, and no amount of slot bookkeeping supplies it. Nor does the passive character of the slot transport create information: moving labels over fixed scores can re-level coverage, but the conditional sharpness of the resulting intervals is fixed by the scores themselves — the information-gap point made elsewhere on this site (Cotton). A natural question is whether an *active* dual exists: a herding-like procedure in index space that chooses which calibration slots to retain or reweight, steering conditional coverage the way herding steers feature means. Localized and weighted conformal methods can be read as first steps in that direction, and the same information-theoretic ceiling applies to them. We leave the speculation at one sentence.

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